

Pasos para Docker

Reviso los contenedores que tengo encendidos. Como ya habíamos descargado el SQL en la clase anterior, no es necesario descargar la imagen.

```
C:\Users\arago>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
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En esta imagen podemos ver los contenedores disponibles apagados. Ahí se encuentra SQL.

```
C:\Users\arago>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e118fcfcb475	mysql:8	"docker-entrypoint.s..."	3 days ago	Exited (255) 40 minutes ago	0.0.0.0:3306->3306/tcp, 33060/tcp	mi-mysql
7bc591a7ea75	hello-world	"/hello"	3 days ago	Exited (0) 3 days ago		affectionate_carson

Prendemos el contenedor y buscamos en la lista los contenedores encendidos, en este caso, SQL está encendido.

Tambien puedes usar Docker start my-mysql

```
C:\Users\arago>docker start e118fcfcb475
e118fcfcb475
```

```
C:\Users\arago>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e118fcfcb475	mysql:8	"docker-entrypoint.s..."	3 days ago	Up 3 seconds	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp	mi-mysql

Ingresamos al contenedor de SQL

```
C:\Users\arago>docker exec -it e118fcfcb475 bash
bash-5.1# |
```

Nos logreamos dentro de MySQL usando mysql -u root -p

Password: root

```
C:\Users\arago>docker exec -it e118fcfcb475 bash
bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.11 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Entramos y usamos la base de datos library

```
mysql> CREATE DATABASE library;
Query OK, 1 row affected (0.01 sec)

mysql> USE library;
Database changed
mysql> |
```

Creamos las tablas users, bookLoan y books

```
mysql> CREATE TABLE users (
  ->     id INT AUTO_INCREMENT,
  ->     name VARCHAR(100) NOT NULL,
  ->     user VARCHAR(50),
  ->     email VARCHAR(100) UNIQUE NOT NULL,
  ->     PRIMARY KEY (id)
  -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> CREATE TABLE books (
  ->     id INT AUTO_INCREMENT,
  ->     title VARCHAR(150) NOT NULL,
  ->     author VARCHAR(100) NOT NULL,
  ->     PRIMARY KEY (id)
  -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE bookLoan (
  ->     id_loan INT AUTO_INCREMENT,
  ->     dateLoan DATE NOT NULL,
  ->     user_id INT NOT NULL,
  ->     book_id INT NOT NULL,
  ->     PRIMARY KEY (id_loan),
  ->     CONSTRAINT fk_loan_user FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE ON UPDATE CASCADE,
  ->     CONSTRAINT fk_loan_book FOREIGN KEY (book_id) REFERENCES books(id) ON DELETE CASCADE ON UPDATE CASCADE
  -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> show tables;
+-----+
| Tables_in_library |
+-----+
| bookLoan          |
| books             |
| users             |
+-----+
3 rows in set (0.00 sec)
```

Inserto registros en las tablas

```
mysql> INSERT INTO users (name, user, email) VALUES
-> ('Ana Gmez', 'anag', 'ana@email.com'),
-> ('Carlos Ruz', 'carlosr', 'carlos@email.com'),
-> ('Mara Lpez', 'marial', 'maria@email.com');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO books (title, author) VALUES
-> ('El Principito', 'Antoine de Saint-Exupry'),
-> ('Cien Aos de Soledad', 'Gabriel Garca Mrquez'),
-> ('Don Quijote de la Mancha', 'Miguel de Cervantes');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO bookLoan (dateLoan, user_id, book_id) VALUES
-> ('2026-03-01', 1, 1),
-> ('2026-03-02', 2, 2),
-> ('2026-03-05', 1, 2);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Se muestran los datos registrados en las tablas

```
mysql> select * from users;
+----+-----+-----+-----+
| id | name      | user    | email                |
+----+-----+-----+-----+
| 1  | Ana Gmez  | anag    | ana@email.com        |
| 2  | Carlos Ruz | carlosr | carlos@email.com     |
| 3  | Mara Lpez | marial  | maria@email.com      |
+----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from books;
+----+-----+-----+
| id | title                | author                |
+----+-----+-----+
| 1  | El Principito        | Antoine de Saint-Exupry |
| 2  | Cien Aos de Soledad  | Gabriel Garca Mrquez  |
| 3  | Don Quijote de la Mancha | Miguel de Cervantes   |
+----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from bookLoan;
+-----+-----+-----+-----+
| id_loan | dateLoan | user_id | book_id |
+-----+-----+-----+-----+
| 1       | 2026-03-01 | 1       | 1       |
| 2       | 2026-03-02 | 2       | 2       |
| 3       | 2026-03-05 | 1       | 2       |
+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```